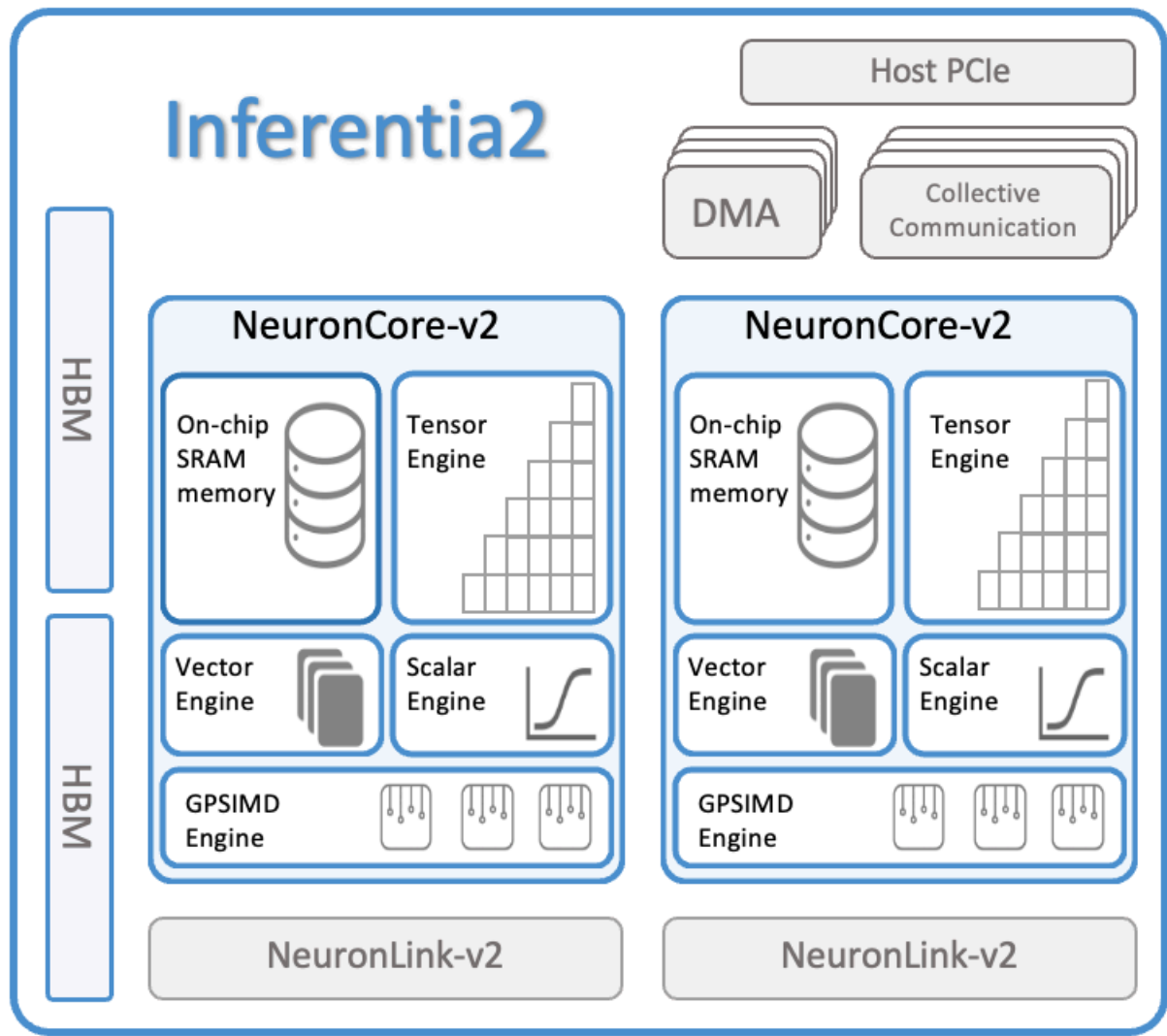


Inferentia2 Architecture

This document is relevant for: Inf2

At the heart of each Inf2 instance are up to twelve Inferentia2 chips (each with two [NeuronCore-v2](#) cores). Inferentia2 is the second generation AWS purpose-built Machine Learning inference accelerator. The Inferentia2 chip architecture is depicted below:



Each Inferentia2 chip consists of:

Compute	Two NeuronCore-v2 cores, delivering 380 INT8 TOPS, 190 FP16/BF16/cFP8/TF32 TFLOPS, and 47.5 FP32 TFLOPS.
Device Memory	32GiB of high-bandwidth device memor (HBM) (for storing model state), with 820 GiB/sec of bandwidth.
Data Movement	1 TB/sec of DMA bandwidth, with inline memory compression/decompression.
NeuronLink	NeuronLink-v2 for chip-to-chip interconnect enables high-performance collective compute for co-optimization of latency and throughput.
Programmability	Inferentia2 supports dynamic shapes and control flow, via ISA extensions of NeuronCore-v2 and custom-operators via the deeply embedded GPSIMD engines.

For a more detailed description of all the hardware engines, see [NeuronCore-v2](#).

[Skip to main content](#)

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